



University
of Exeter

COMM113

Deep Learning

My research: **A quick intro about me!**

- I study **how to develop trustworthy deep learning models.**
- In practice, that means how to improve DL performance in **new environments** and **adversarial environments**

What this course is about

- Core concepts of Deep Learning (DL),
- How to improve generalisation and robustness of DL models
- Strengths and weaknesses of different methods
- All at a conceptual level... mostly

What this course is about

Backpropagation

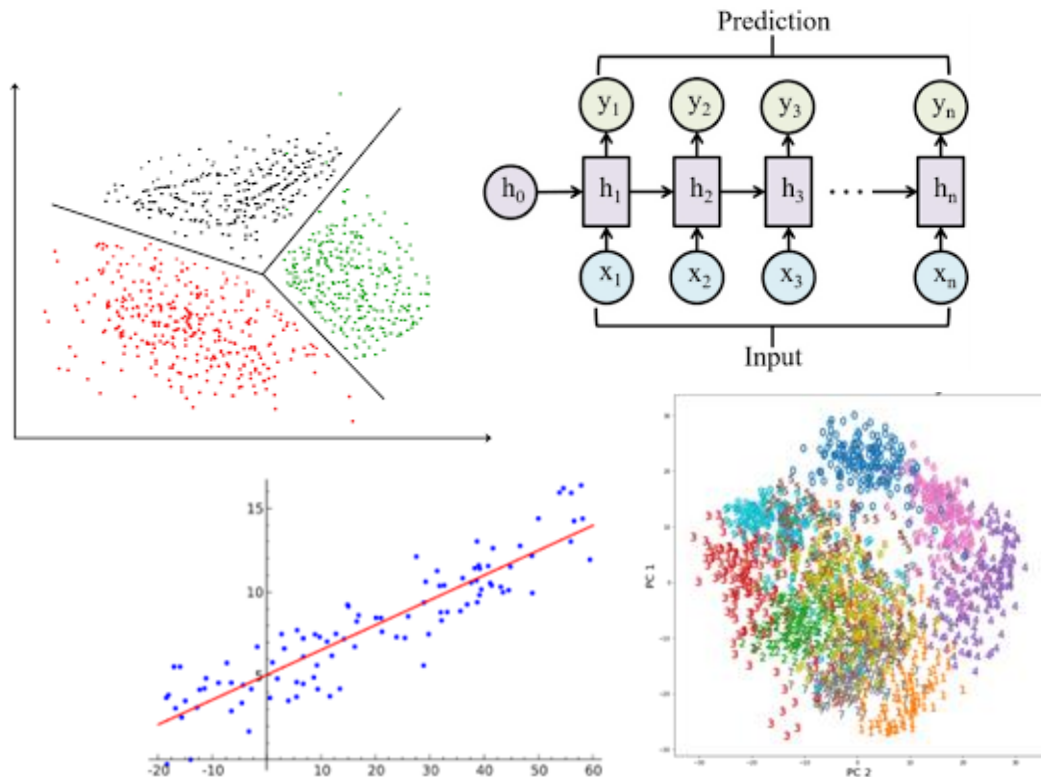
Regularisations

CNN

Object Detection

RNN

Transformer



Assessment:

- Two assessments:
 - Coursework 1, worth 30% of your mark due 17 November 2025
(MLP on FashionMNIST using Pytorch (30%) and Numpy-only (70%))
 - Written Exam, worth 70% of your mark
(Conceptual: 60%, Computation: 40%, e.g., backpropagation)
 - More detail to come!

Lectures:

- Lectures:
 - Fridays 10:35 AM – 11:35 AM
LT4.2 /Queens
 - Fridays 4:35 PM – 5:35 PM
LT4.2 /Queens
 - There might be changes. Check your timetable.
- Recorded and posted on ele.exeter.ac.uk

The COMM113 teaching team

Lecturers:



me!

Gaojie (Jay) Jin

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Teaching Assistant (TA):

Me!

Practical workshops:

- Optional, but **very** useful
- You can work in groups and get help from the teaching assistants
 - Monday 10:35 AM-11:35 AM, 207 /Harrison

The normal workshop will start in week 2.

- TA: Jay

Resources

Textbooks (Foundational & Practical)

- Deep Learning (Goodfellow, Bengio, Courville, 2016) – classic reference for theory & foundations. <https://www.deeplearningbook.org/>
- Dive into Deep Learning (Zhang et al., 2023, open-source) – interactive book with code in PyTorch/MXNet. <https://d2l.ai>
- Neural Networks and Deep Learning (Michael Nielsen, free online book) – beginner-friendly introduction. <http://neuralnetworksanddeeplearning.com>

Framework-Specific (PyTorch Focused)

- PyTorch Official Tutorials – hands-on beginner to advanced guides. <https://pytorch.org/tutorials>
- Deep Learning with PyTorch (Eli Stevens, Luca Antiga, Thomas Viehmann, 2020) – practical book with PyTorch focus.
- PyTorch Lightning Docs – structured PyTorch training framework. <https://lightning.ai/docs>

Online Courses

- CS231n: Convolutional Neural Networks for Visual Recognition (Stanford) – highly recommended for computer vision. <http://cs231n.stanford.edu>
- Deep Learning Specialization (Andrew Ng, Coursera) – covers fundamentals, practical coding.
- Fast.ai Practical Deep Learning for Coders – very hands-on, PyTorch-based. <https://course.fast.ai>

Outline

Week	Date	Topic
week 1	26-Sep	Intro to Deep Learning
week 2	03-Oct	Neural Networks and Backpropagation
week 3	10-Oct	Regularisations I
week 4	17-Oct	Regularisations II
week 5	24-Oct	Convolutional Neural Networks for Image Classification
week 6	31-Oct	Deep Learning for Object Detection and Image Segmentation
week 7	07-Nov	Recurrent Neural Network
week 8	14-Nov	Transformer and Language Models I
week 9	21-Nov	Transformer and Language Models II
week 10	28-Nov	Robustness and Trustworthy Deep Learning I
week 11	05-Dec	Robustness and Trustworthy Deep Learning II
week 12	12-Dec	Revision Lectures

Thank you!

If you want to ask us any questions about the module, the university, or anything else, drop us an email.

Pro tip: write **COMM113** on the email subject line, to make sure we give it priority.



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